

Julian Karlbauer M.Sc.

Systems-focused Research Engineer

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Specializing in Human-Computer-Interaction, Machine Learning, and Extended Reality (XR) systems. Expert in developing and deploying research prototypes into real-world applications, evidenced by work at Siemens, Audi, and Silicon Valley startups. Experienced HCI researcher with publications in leading venues including CHI, ToCHI, and MUM.

EXPERIENCE

	Sabbatical	Latin America, Middle East, Europe, Asia
	<i>Travelling and Volunteering</i>	06/2025 – 12/2025
	A selection of my experiences includes: (i) contributing to journalism in the Middle East resulting in two published newspaper articles, (ii) extensive endurance and alpine trekking, (iii) silent Theravada Buddhist retreat, (iv) farm work.	
	Phygtl Inc.	Palo Alto, USA (<i>part-time, remote</i>)
	<i>Founding Engineer</i>	12/2024 – 06/2025
	Equity-based founding role. Worked on a multiplayer social AR platform enabling GenAI based 3D co-creation. Responsible for frontend implementation, architectural design, code quality, and Android/iOS releases. Collaborated directly with the executive team to define the technical roadmap.	
	Self Employed	Ulm, Germany (<i>part-time, remote</i>)
	<i>Independent Consultant for XR Systems</i>	11/2024 – 04/2025
	Led the AR development of a decentralized emergency simulation prototype targeted at Apple Vision Pro for an industrial client. The prototype was demoed at the GISEC Global 2025 exhibition.	
	Ulm University	Ulm, Germany
	<i>Postgraduate HCI Researcher</i>	01/2024 – 10/2024
	Held a 4-year Research Associate contract under Prof. Dr. Enrico Rukzio fully funded by the German Research Foundation (DFG). I conducted research on Generative-AI supported 3D authoring in AR to enhance digital manufacturing workflows. I demoed my work at the ACM SCF24 Conference. I co-authored a paper [1] at the ACM MUM24 conference. Discontinued to pursue personal goals.	
	Siemens AG	Munich, Germany
	<i>Student Researcher</i>	04/2023 – 11/2023
	Built an impactful AR prototype that caused traction in the research group and was recognized as a future candidate to be demoed to Roland Busch (CEO), a demo at Formnext (leading convention on Additive Manufacturing), and development into a commercial Siemens product. The system is an important step towards usable Digital Twin for industrial metal 3D printing. It enables anomaly detection, 3D visualization, data labelling, and data-lake integration to enhance User Experience in industrial scale Additive Manufacturing	
	Audi Production Lab	Ingolstadt, Germany
	<i>Software Engineering Intern</i>	03/2022 – 09/2022
	Worked in an engineering team to develop a mobile cross-platform AR application designed for Audi internal communication. My focus was on Unity3D based iOS development and interaction design.	
	TriCAT GmbH	Ulm, Germany
	<i>R&D Intern</i>	03/2021 – 06/2021
	Worked independently on an AR system on HoloLens 2 for in-situ real-time object recognition and 3D localization using CNNs and environment meshes with a compute efficient remote GPU architecture to tackle unsolved customer challenges. Through my work I pioneered the use of AI tools in the company.	
	Ulm University Department of Media Informatics	Ulm, Germany
	<i>Research Assistant</i>	04/2018 – 03/2021
	Co-authored papers at CHI [2] and ToCHI [3]. Developed VR Research Prototypes, conducted User Studies, and tutored students.	

EDUCATION

	Ulm University	Ulm, Germany
	M. Sc. in Media Informatics	10/2020 – 03/2024
	Thesis Title: <i>Show, Don't Tell: Enhancing Quality Assurance in Additive Manufacturing through Augmented Reality (Grade 1.0)</i>	
	Norwegian University of Science and Technology	Trondheim, Norway
	Visiting Student in Generative AI	08/2021 – 12/2021
	Thesis Title: <i>Raw Audio Music Emotion Composition using Mood Annotated Spotify Playlists (Grade 1.0)</i>	
	Ulm University	Ulm, Germany
	B. Sc. in Media Informatics	04/2016 – 07/2020
	Thesis Title: <i>An Exploration of Foveated Blue Light Filters in Virtual Reality (Grade 1.0)</i>	

ACHIEVEMENTS, VOLUNTEERING, SERVICES

- Completed over 1,000km of solo high-altitude and long-distance trekking, including 500km of the Camino de Santiago (21 days) and self-supported navigation of the Peruvian Andes (> 5,000m altitude).
- Volunteered on a private farm in Thailand. My tasks included picking and processing of coffee, general property maintenance, and electronics repairs.
- Participated in a 2-week silent Theravada Buddhist Monastery retreat consisting of extensive daily meditation, property maintenance, and participating in traditional monastic life.
- Accompanied a journalist in the Middle East resulting in two published newspaper articles. I was responsible for capturing footage, assisting in interviews, and mediating interviewing partners.
- Solo performances of self-composed pieces on classical guitar as part of classical concerts.
- Conference Reviewer CHI 2025
- Served as a student volunteer at the ACM SCF 2024 conference.
- Former long-term member of the Reger Vokalensemble in Ulm

PUBLICATIONS

- [1] Tobias Drey, Nico Rixen, **Julian Karlbauer**, and Enrico Rukzio. 2024. VRCreatIn: Taking In-Situ Pen and Tablet Interaction Beyond Ideation to 3D Modeling Lighting and Texturing. In Proceedings of the International **Conference on Mobile and Ubiquitous Multimedia (MUM '24)**. Association for Computing Machinery, New York, NY, USA, 24–35. <https://doi.org/10.1145/3701571.3701580>
- [2] Teresa Hirzle, Fabian Fischbach, **Julian Karlbauer**, Pascal Jansen, Jan Gugenheimer, Enrico Rukzio, and Andreas Bulling. 2022. Understanding, Addressing, and Analysing Digital Eye Strain in Virtual Reality Head-Mounted Displays. **ACM Trans. Comput.-Hum. Interact.** 29, 4, Article 33 (August 2022), 80 pages. <https://doi.org/10.1145/3492802>
- [3] Tobias Drey, Jan Gugenheimer, **Julian Karlbauer**, Maximilian Milo, and Enrico Rukzio. 2020. VRSketchIn: Exploring the Design Space of Pen and Tablet Interaction for 3D Sketching in Virtual Reality. In Proceedings of the 2020 **CHI Conference on Human Factors in Computing Systems (CHI '20)**. Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376628>